

BUSINESS PROBLEM

A leading retail company is experiencing shifts in customer purchasing behavior across demographics, product categories, and sales channels (online vs. offline). To remain competitive, the company needs to better understand the key drivers influencing customer decisions and long-term engagement.

The objective of this project is to analyze consumer shopping data to uncover actionable insights that can improve **sales performance, customer satisfaction, and retention strategies**.

❖ Key Business Question

How can consumer shopping data be leveraged to identify behavioral trends, enhance customer engagement, and optimize marketing and product strategies?

❖ Objectives

- Identify key factors influencing purchasing decisions (e.g., discounts, reviews, seasonality).
- Analyze customer segments and buying patterns.
- Understand drivers of repeat purchases and customer loyalty.

❖ Project Approach

1. Data Preparation (Python): Data cleaning, transformation of raw data for DA.
2. Data Analysis (SQL): Query-based analysis for customer segmentation, trends, purchase drivers and loyalty.
3. Visualization (Power BI): Interactive dashboard for business insights.
4. Reporting: Clear communication of insights and recommendations.
5. GitHub Repository: Includes all project files such as Python scripts, SQL queries, dashboards, and reports in a well-structured repository.

❖ Expected Outcome

- Actionable insights into customer behavior.
- Data-driven recommendations for marketing and sales strategies.
- Improved decision-making through interactive dashboards.